

# Lecture 7: Asynchronous programming

Presentations students, introduction to async

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**DevLab Rust 2025**

Tuesday December 9, 2025

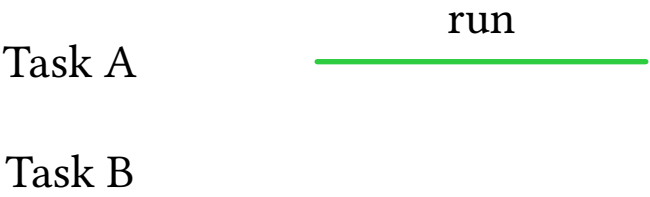
[github.com/wvhulle/rust-course-ghent](https://github.com/wvhulle/rust-course-ghent)

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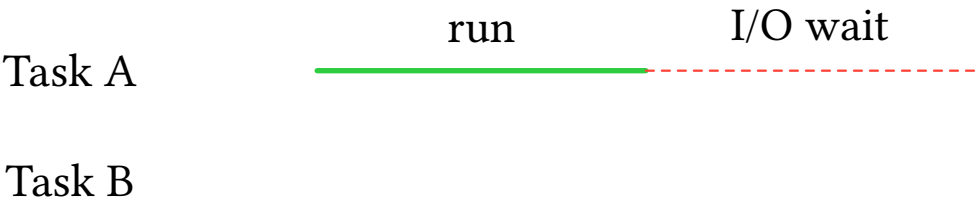
# 2.1. Introduction

Multiple tasks execute concurrently by running until they would block on I/O, then yielding to another ready task.



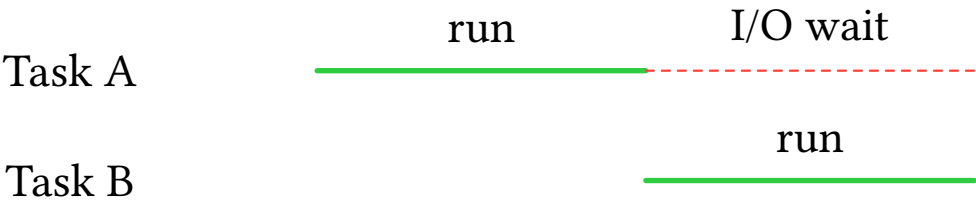
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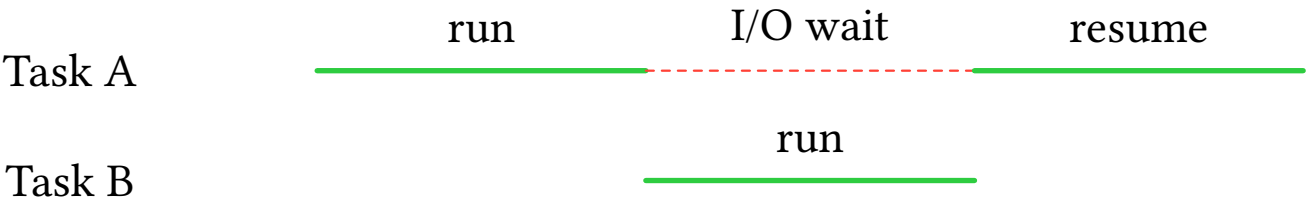
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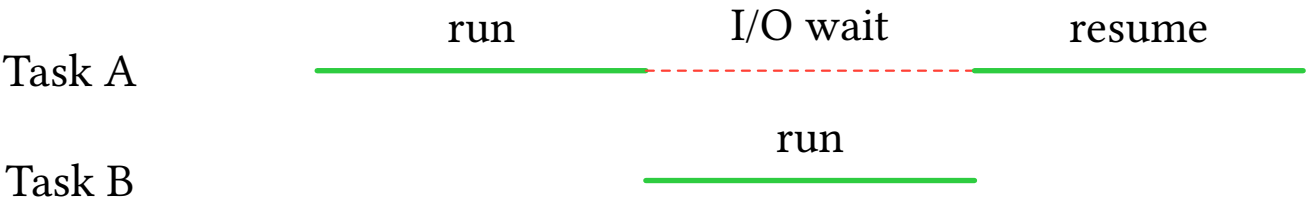
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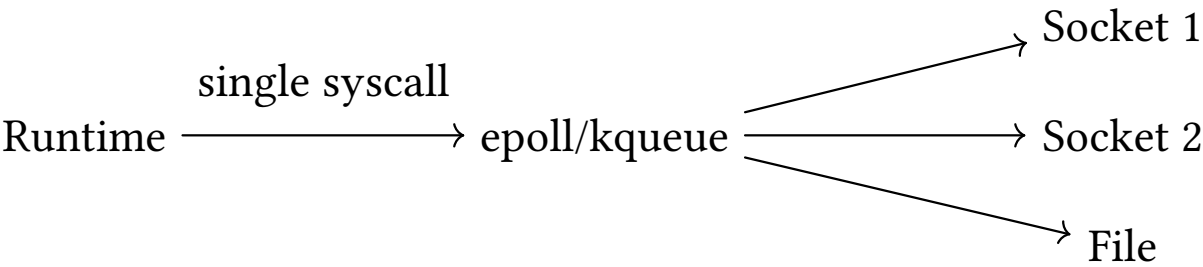


# 2.1. Introduction

Multiple tasks execute concurrently by running until they would block on I/O, then yielding to another ready task.



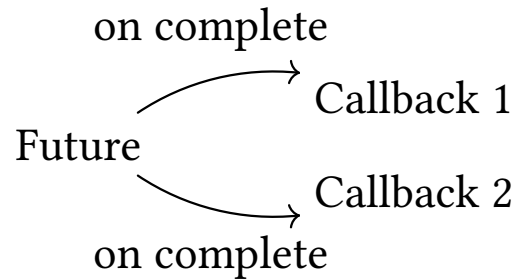
Per-task overhead is minimal because operating systems provide primitives (epoll, kqueue, IOCP) that monitor many I/O operations with a single system call.





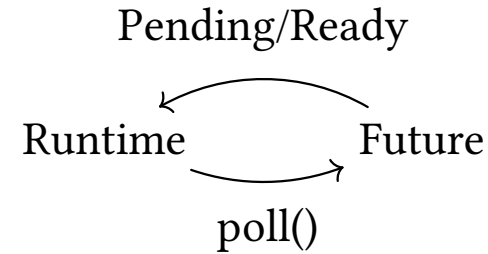
## 2.2. Comparison with Python

### Python asyncio



- Callback-based model
- Register callbacks
- Built-in event loop
- Interpreter overhead

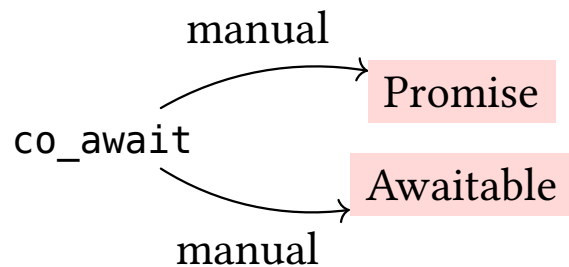
### Rust async/await



- Poll-based model
- Runtime polls futures
- External runtimes (tokio, async-std)
- Zero-cost state machines

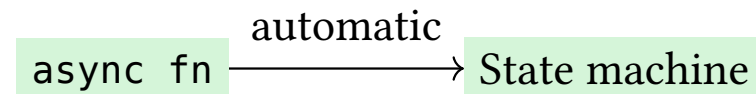
## 2.3. Comparison with C++

### C++20 Coroutines



- Manual promise types
- Manual awaitable objects
- No standard runtime
- **Heap allocation** by default

### Rust async/await



- Compiler generates state machines
- Built-in syntax
- Mature ecosystem (tokio)
- **Stack-allocated** by default

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### 3.1. Await syntax

At a high level, async Rust code looks very much like “normal” sequential code:

```
1  use futures::executor::block_on;
2
3  async fn count_to(count: i32) {
4      for i in 0..count {
5          println!("Count is: {i}!");
6      }
7  }
8
9  async fn async_main(count: i32) {
10     count_to(count).await;
11 }
12
13 fn main() {
14     block_on(async_main(10));
15 }
```

*([playground link](#))*

## 3.2. Futures

Future is a trait, implemented by objects that represent an operation that may not be complete yet. A future can be polled, and poll returns a Poll.

```
1  use std::pin::Pin; use std::task::Context;
2
3  pub trait Future {
4      type Output;
5      fn poll(self: Pin<&mut Self>, cx: &mut Context<'_>) -> Poll<Self::Output>;
6  }
7
8  pub enum Poll<T> {
9      Ready(T),
10     Pending,
11 }
```

[\(playground link\)](#)

What is the role of Context?

## 3.2. Futures

Future is a trait, implemented by objects that represent an operation that may not be complete yet. A future can be polled, and poll returns a Poll.

```
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4     type Output;
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6 }
7
8 pub enum Poll<T> {
9     Ready(T),
10    Pending,
11 }
```

*([playground link](#))*

What is the role of Context?

Context allows a Future to schedule itself to be polled again when an event such as a timeout occurs.

### 3.3. JoinHandle

One common type implementing Future is a Tokio JoinHandle:

```
1 use tokio::task;  
2  
3 let join = task::spawn(async {  
4     panic!("something bad happened!");  
5 });  
6  
7 // The returned result indicates that the task failed.  
8 assert!(join.await.is_err());
```

[\*\(playground link\)\*](#)

|           |                                            |           |
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## 4.1. State machine

Rust transforms async functions into state machines that implement Future.

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Key characteristics:

- Calling an async function constructs and returns a future
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- Tracks progress through suspension points

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Rust transforms async functions into state machines that implement `Future`.

Key characteristics:

- Calling an async function constructs and returns a future
- Does not execute immediately (lazy)
- Tracks progress through suspension points

The state machines are allocated on the stack by default. When the state machine transitions to the next state, the async runtime does not move it.

## 4.2. Transformation

Consider this async function:

```
1 async fn two_d10(modifier: u32) -> u32 {  
2     let first_roll = roll_d10().await;  
3     let second_roll = roll_d10().await;  
4     first_roll + second_roll + modifier  
5 }
```

*[\(playground link\)](#)*

## 4.2. Transformation

Consider this async function:

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1 async fn two_d10(modifier: u32) -> u32 {  
2     let first_roll = roll_d10().await;  
3     let second_roll = roll_d10().await;  
4     first_roll + second_roll + modifier  
5 }
```

*[\(playground link\)](#)*

The compiler generates an enum that:

- Tracks each suspension point (each `.await`)
- Stores all live variables needed at that state
- Implements the Future trait

### 4.3. State machine structure

The generated enum looks conceptually like this:

```
1 enum TwoDiceFuture {  
2     Start { modifier: u32 },  
3     FirstRoll { modifier: u32, first_roll: impl Future<Output = u32> },  
4     SecondRoll { modifier: u32, first_result: u32, second_roll: impl Future<Output =  
5         u32> },  
6     Done,  
7 }
```

[\(playground link\)](#)

### 4.3. State machine structure

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5         u32> },  
6     Done,  
}
```

[\(playground link\)](#)

Each state stores:

- The variables still needed
- The future being awaited (if any)

## 4.4. Recursion

Deeply nested async functions create large compiler-generated Future types.

Each function's Future contains its callees' Futures.



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Deeply nested async functions create large compiler-generated Future types.

Each function's Future contains its callees' Futures.

Recursive async functions require boxing to avoid infinite type sizes:

```
1  async fn count_to(n: u32) {  
2      if n > 0 {  
3          Box::pin(count_to(n - 1)).await;  
4          println!("{n}");  
5      }  
6  }  
7  
8  Box::pin(count_to(n - 1)).await;
```

*([playground link](#))*

## 4.4. Recursion

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Recursive async functions require boxing to avoid infinite type sizes:

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5     }  
6 }  
7  
8 Box::pin(count_to(n - 1)).await;
```

*([playground link](#))*

This adds heap allocation overhead but makes recursion possible.

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## 5.1. Runtimes

A runtime provides support for:

- performing operations asynchronously (a **reactor**)
- and is responsible for executing futures (an **executor**).

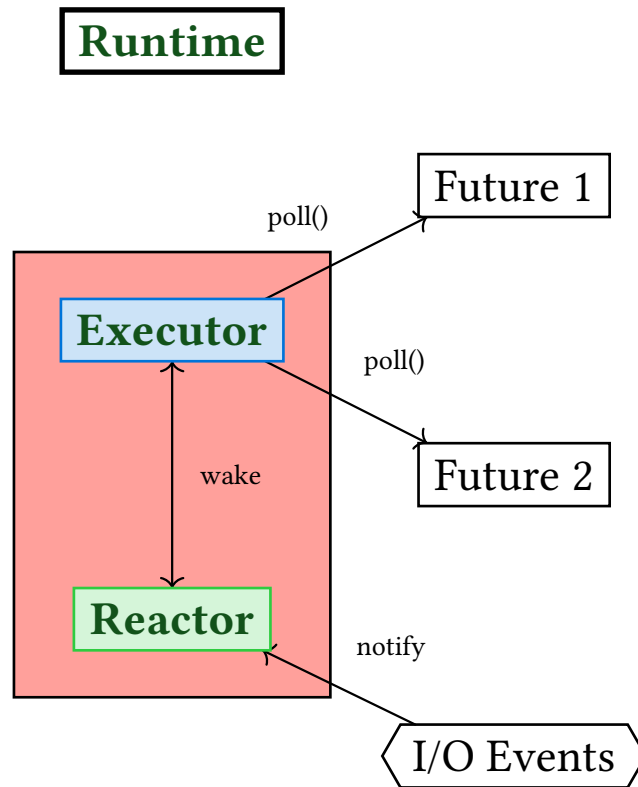
## 5.1. Runtimes

A runtime provides support for:

- performing operations asynchronously (a **reactor**)
- and is responsible for executing futures (an **executor**).

Rust does not have a “built-in” runtime, but several options are available:

- futures offers a bare bones executor ThreadPool
- tokio offers a Runtime with reactor included (supplies “time” etc.)



```
1  use tokio::time;
2  async fn count_to(count: i32) {
3      for i in 0..count {
4          println!("Count in task: {i}!");
5          time::sleep(time::Duration::from_millis(5)).await;
6      }
7  }
8  #[tokio::main]
9  async fn main() {
10     tokio::spawn(count_to(10));
11     for i in 0..5 {
12         println!("Main task: {i}");
13         time::sleep(time::Duration::from_millis(5)).await;
14     }
15 }
```

*([playground link](#))*

Async tasks are also aborted if not awaited in main.

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## 6.1. Tasks

## 6. Common async datatypes and functions

A task is a top-level future that may be sent to different worker threads (if necessary).

```
1  #[tokio::main]
2  async fn main() -> io::Result<()> {
3      let listener = TcpListener::bind("127.0.0.1:0").await?;
4
5      loop {
6          let (mut socket, addr) = listener.accept().await?;
7
8          tokio::spawn(async move {
9              // Handle connection
10             });
11     }
12 }
```

*[\(playground link\)](#)*



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9              // Handle connection
10             });
11     }
12 }
```

*([playground link](#))*

### Warning

Although we use tokio in this section, the same functionality is available in most async runtimes.

Inside `tokio::spawn`:

```
1 socket.write_all(b"Who are you?\n").await?;  
2  
3 let mut buf = vec![0; 1024];  
4 let name_size = socket.read(&mut buf).await?;  
5 let name = std::str::from_utf8(&buf[..name_size])?.trim();  
6  
7 socket.write_all(format!("Thanks, {name}!\n").as_bytes()).await?;    (playground link)
```

### Info

An async block like `async move {}` is like a scope block that may contain awaits.

Refactor `session7/examples/tasks.rs`:

- Create async helper function from async block
- Improve error handling

```
1 async fn ping_handler(mut input: mpsc::Receiver<()>) {  
2     let mut count: usize = 0;  
3     while let Some(_) = input.recv().await {  
4         count += 1;  
5         println!("Received {count} pings so far.");  
6     }  
7 }
```

*[\(playground link\)](#)*

```
1 async fn ping_handler(mut input: mpsc::Receiver<()>) {  
2     let mut count: usize = 0;  
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4         count += 1;  
5         println!("Received {count} pings so far.");  
6     }  
7 }
```

[\(playground link\)](#)

```
1 async fn main() {  
2     let (sender, receiver) = mpsc::channel(32);  
3     let handler_task = tokio::spawn(ping_handler(receiver));  
4     for i in 0..10 {  
5         sender.send(()).await.expect("Failed to send");  
6     }  
7     drop(sender);  
8     handler_task.await.expect("Handler failed");  
9 }
```

[\(playground link\)](#)

We might need to do the same async function multiple times:

```
1 async fn size_of_page(url: &str) -> Result<usize> {  
2     let resp = reqwest::get(url).await?;  
3     Ok(resp.text().await?.len())  
4 }
```

*[\(playground link\)](#)*

How can we await a list of futures concurrently?



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*[\(playground link\)](#)*

How can we await a list of futures concurrently?

We use the `join_all` function.





## 6.4. Joining

We might need to do the same async function multiple times:

```
1 async fn size_of_page(url: &str) -> Result<usize> {  
2     let resp = request::get(url).await?;  
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4 }
```

[\(playground link\)](#)

How can we await a list of futures concurrently?

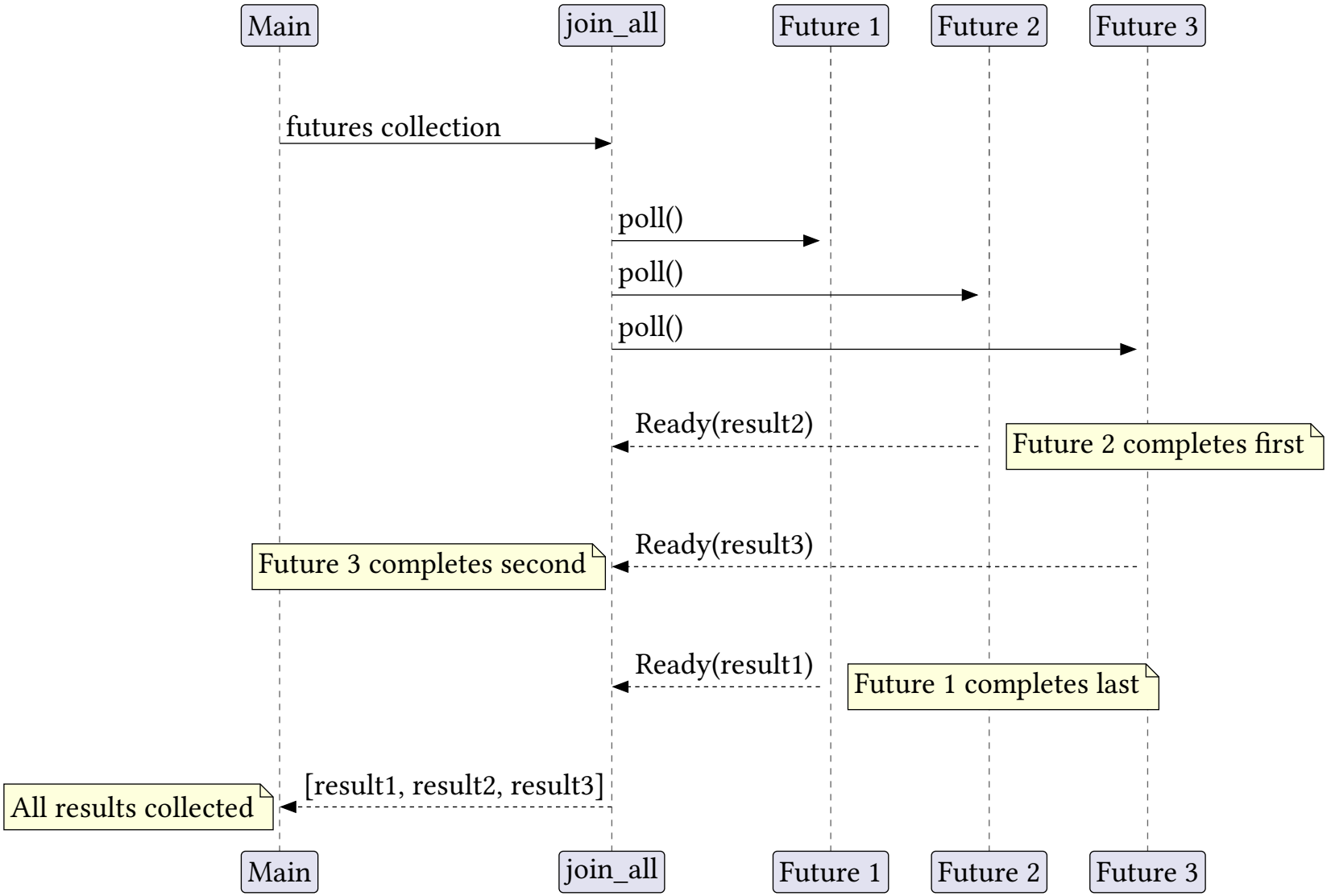
We use the `join_all` function.

```
1 async fn main() {  
2     let urls = ["https://google.com", "https://httpbin.org/ip", /*...*/];  
3     let futures = urls.into_iter().map(size_of_page);  
4     let results = future::join_all(futures).await;  
5     println!("{results:?}");  
6 }
```

[\(playground link\)](#)

pagebreak

# 6.4. Joining



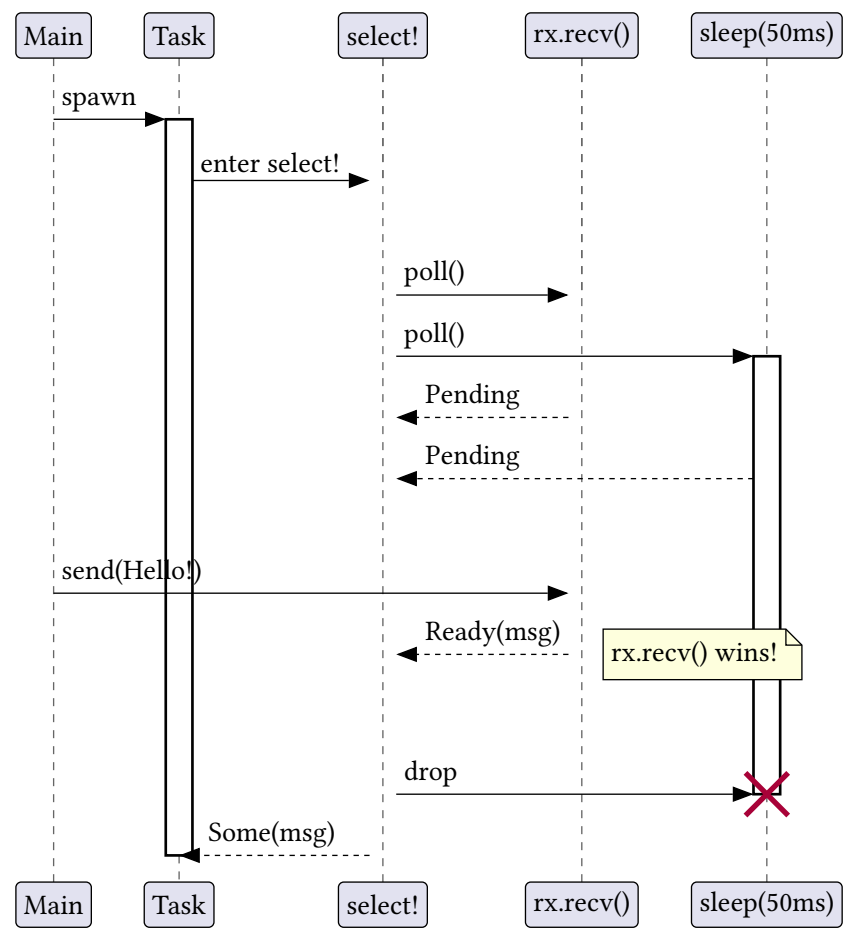
## 6.5. Select

Waits until any of a set of futures is ready. Similar to `Promise.race` or Python's `asyncio.wait()`.

```
1  #[tokio::main]
2  async fn main() {
3      let (tx, mut rx) = mpsc::channel(32);
4
5      tokio::spawn(async move {
6          tokio::select! {
7              Some(msg) = rx.recv() => println!("got: {msg}"),
8              _ = sleep(Duration::from_millis(50)) => println!("timeout"),
9          };
10     });
11
12     sleep(Duration::from_millis(10)).await;
13     tx.send(String::from("Hello!")).await?;
14     // ...
15 }
```

*([playground link](#))*

# 6.5. Select



Streams are runtime-agnostic asynchronous iterators:

```
1  use futures::stream::{self, StreamExt};
2
3  let stream = stream::iter(vec!['a', 'b', 'c']);
4
5  let mut stream = stream.enumerate();
6
7  assert_eq!(stream.next().await, Some((0, 'a')));
8  assert_eq!(stream.next().await, Some((1, 'b')));
9  assert_eq!(stream.next().await, Some((2, 'c')));
10 assert_eq!(stream.next().await, None);
```

*[\(playground link\)](#)*

See my project <https://github.com/wvhulle/clone-stream>

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## 7.1. Blocking executor

CPU blocking tasks will block the executor and prevent other tasks from being executed.

```
1  async fn sleep_ms(start: &Instant, id: u64, duration_ms: u64) {
2      std::thread::sleep(std::time::Duration::from_millis(duration_ms));
3      println!("future {id} finished after {}ms", start.elapsed().as_millis());
4  }
5
6  #[tokio::main(flavor = "current_thread")]
7  async fn main() {
8      let start = Instant::now();
9      let futures = (1..=10).map(|t| sleep_ms(&start, t, t * 10));
10     join_all(futures).await;
11 }
```

*([playground link](#))*



## 7.1. Blocking executor

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*([playground link](#))*

What is a workaround?

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11 }
```

*([playground link](#))*

What is a workaround?

Use async methods: `tokio::time::sleep` instead of `std::thread::sleep`

| Aspect    | Threads (Parallelism)                                         | Async (Concurrency)                                      |
|-----------|---------------------------------------------------------------|----------------------------------------------------------|
| Use case  | CPU-bound tasks: heavy computation                            | I/O-bound tasks: waiting for external resources          |
| Goal      | Utilize multiple CPU cores for true parallelism               | Handle many operations on single thread                  |
| Examples  | Data processing, scientific computing, compilation, rendering | Web servers, database queries, network clients, file I/O |
| Overhead  | Higher: context switching, separate stacks                    | Lower: cooperative yielding, shared stack                |
| When best | Independent computations that can run simultaneously          | Many concurrent operations that mostly wait              |

### 7.3. Deadlocks: Coffman Conditions

A deadlock occurs when four conditions are met simultaneously (Coffman conditions):

| Condition        | Description                                             |
|------------------|---------------------------------------------------------|
| Mutual exclusion | Resources cannot be shared (only one thread at a time)  |
| Hold and wait    | Threads hold resources while waiting for others         |
| No preemption    | Resources cannot be forcibly taken from threads         |
| Circular wait    | Threads form a cycle waiting for each other's resources |

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|------------------|---------------------------------------------------------|
| Mutual exclusion | Resources cannot be shared (only one thread at a time)  |
| Hold and wait    | Threads hold resources while waiting for others         |
| No preemption    | Resources cannot be forcibly taken from threads         |
| Circular wait    | Threads form a cycle waiting for each other's resources |

To prevent deadlock, break at least one condition:

- **Lock ordering:** Always acquire locks in the same order (breaks circular wait)
- **Timeout:** Release locks if waiting too long (breaks hold and wait)
- **Try-lock:** Use `try_lock()` instead of `lock()` (breaks hold and wait)

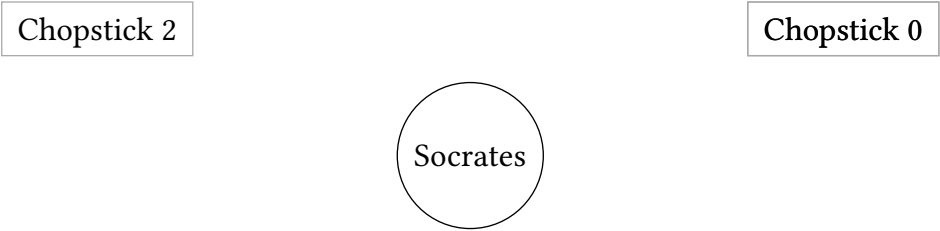
|           |                                            |           |
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8.1. Context

Simplified: 3 philosophers sitting around a table, 3 chopsticks between them.

Rules:

- Each philosopher needs 2 chopsticks (left and right) to eat
- Eating takes time (must hold both chopsticks during eating)
- All philosophers want to eat repeatedly

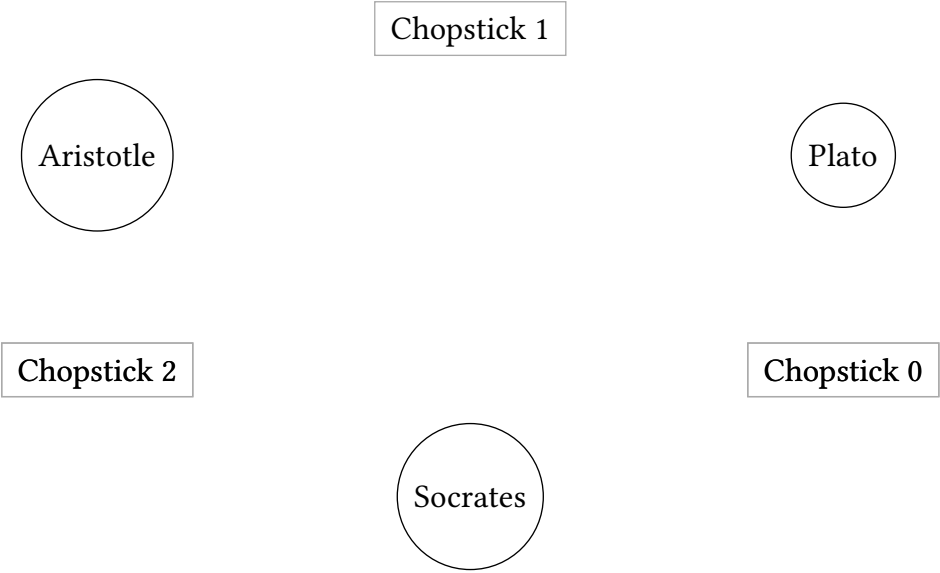


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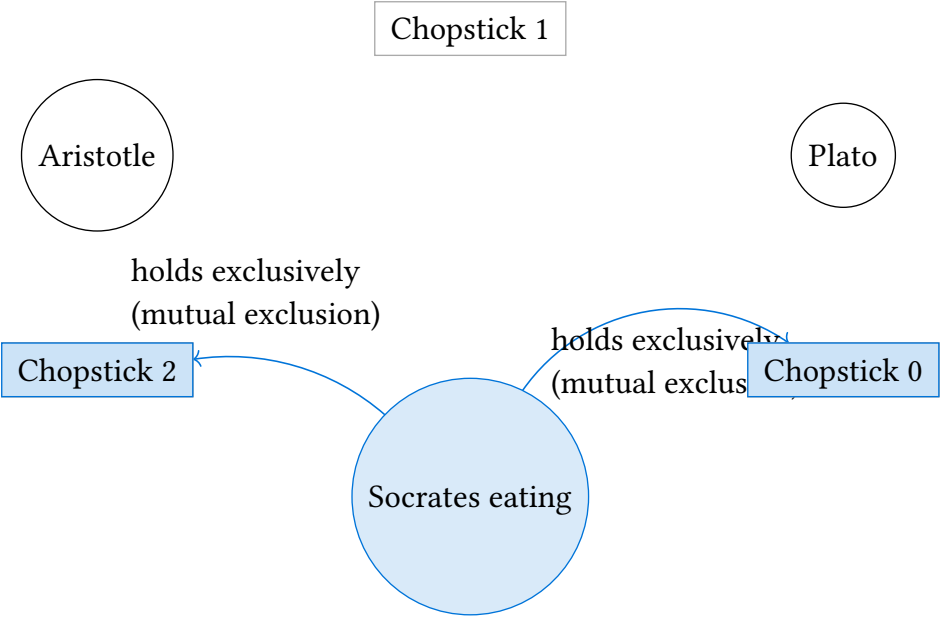


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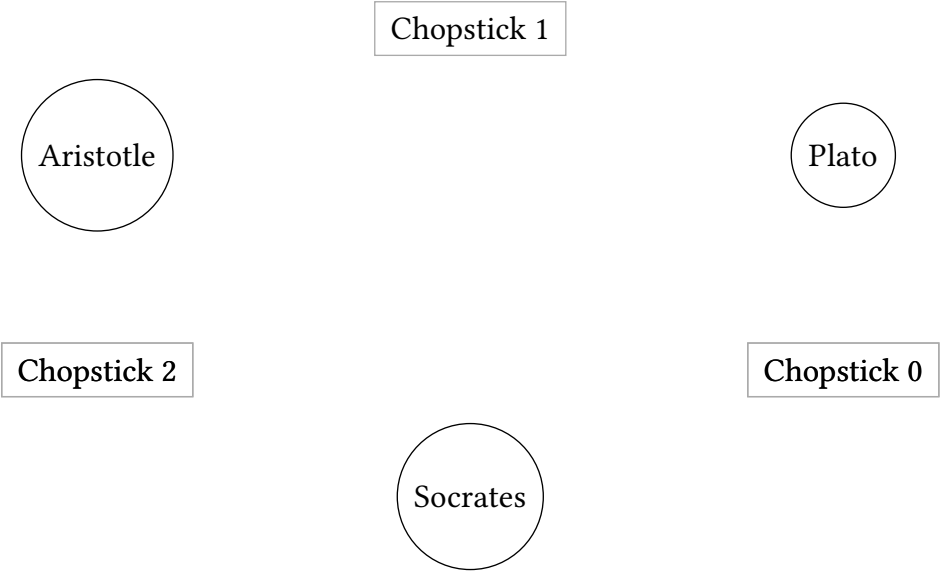


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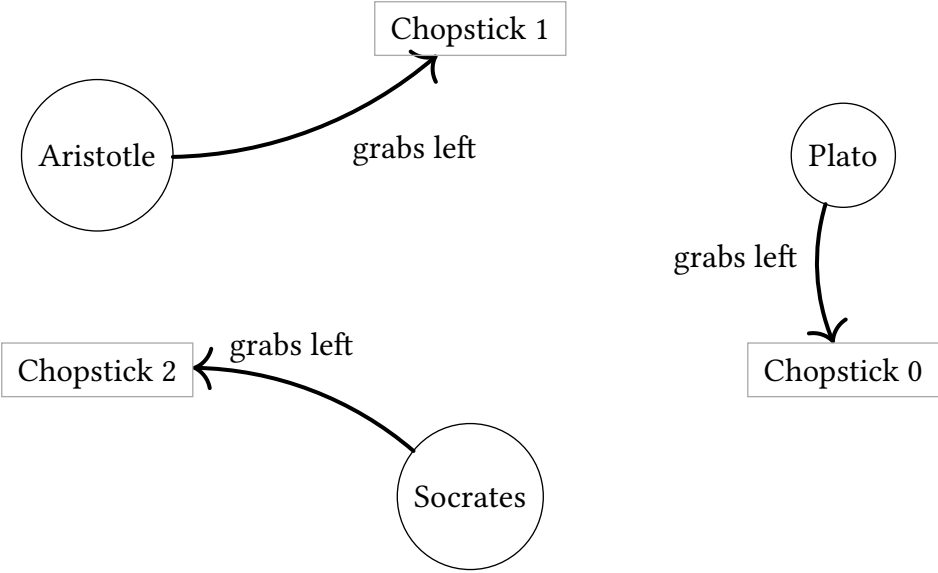


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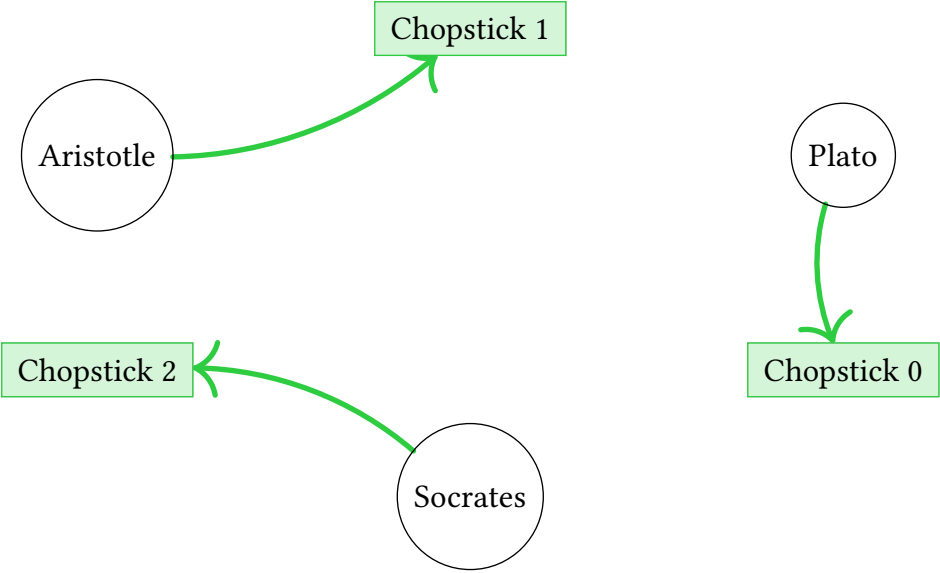


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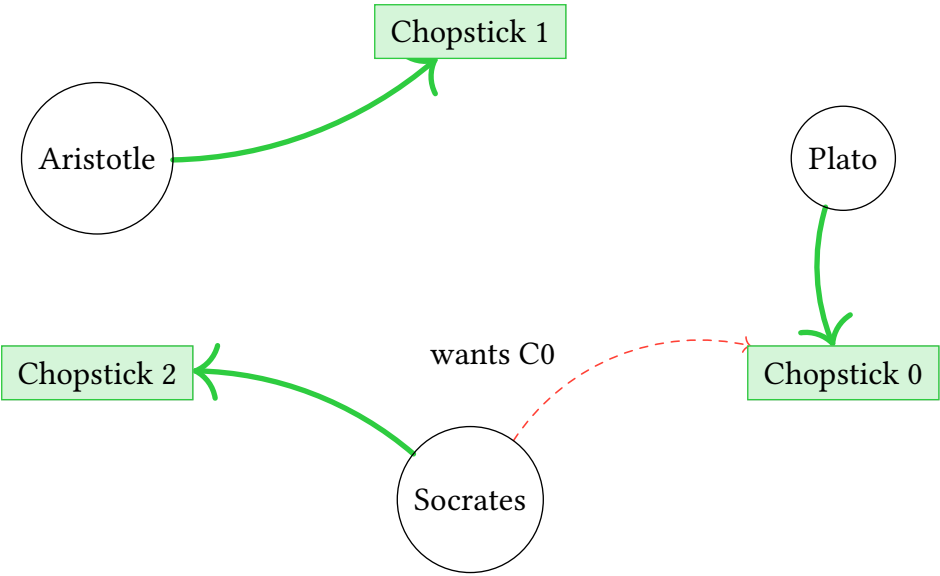


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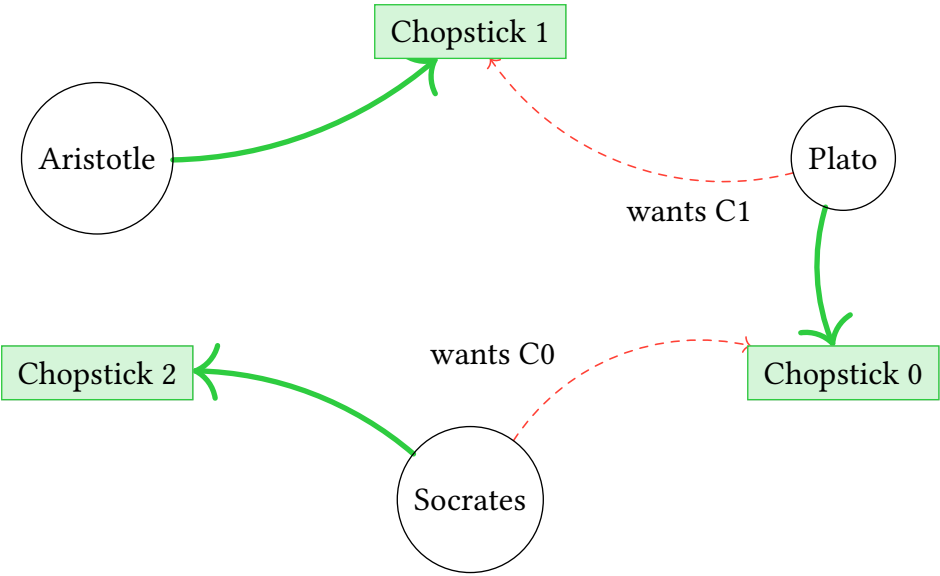


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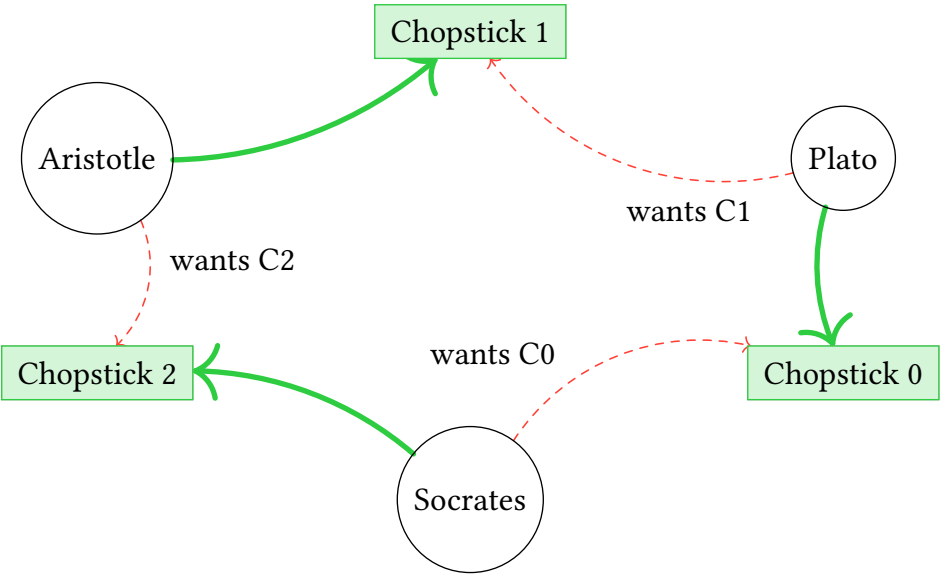


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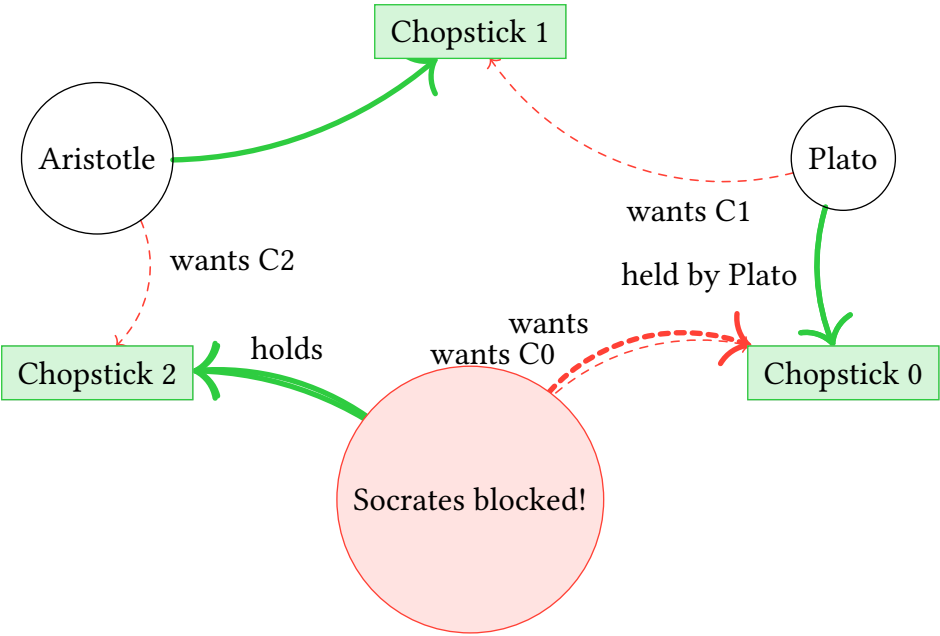


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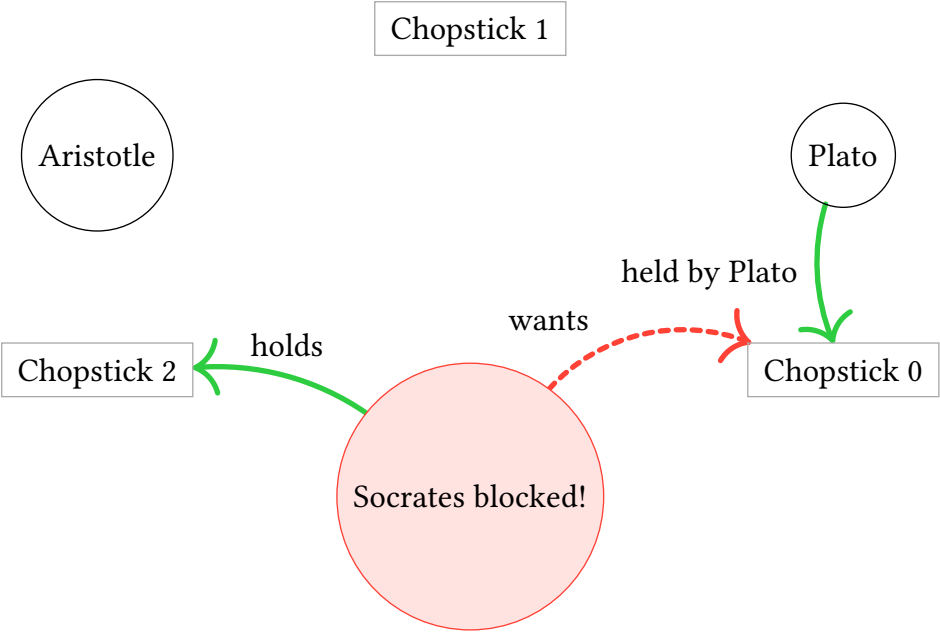


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Legend:

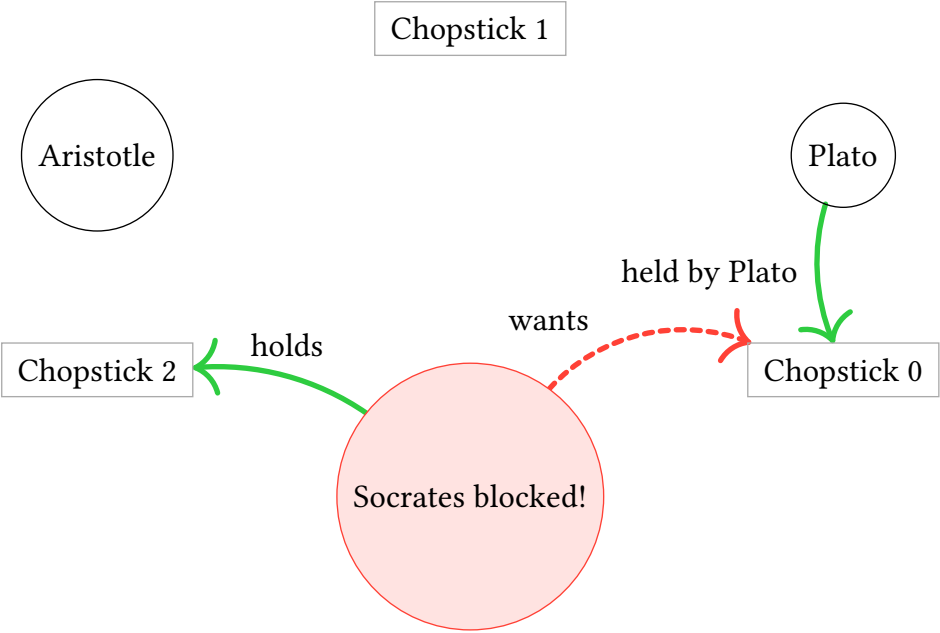
- Trying to grab
- Holding successfully
- - - Wants but blocked

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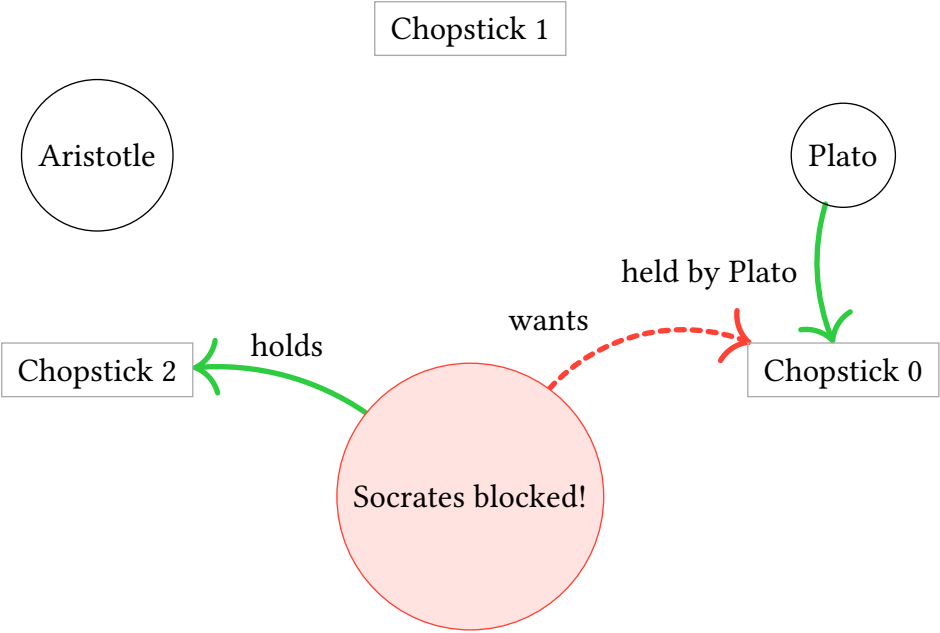
- Holds C2 (green)
- Wants C0 (red dashed)
- But C0 is held by Plato!

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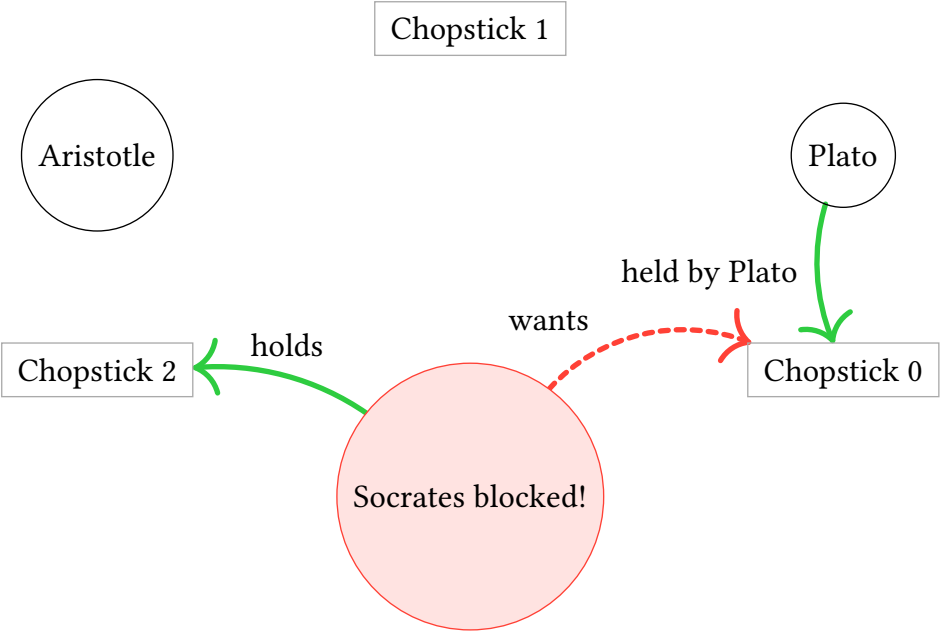
Same for Plato and Aristotle → circular wait → **Deadlock!**

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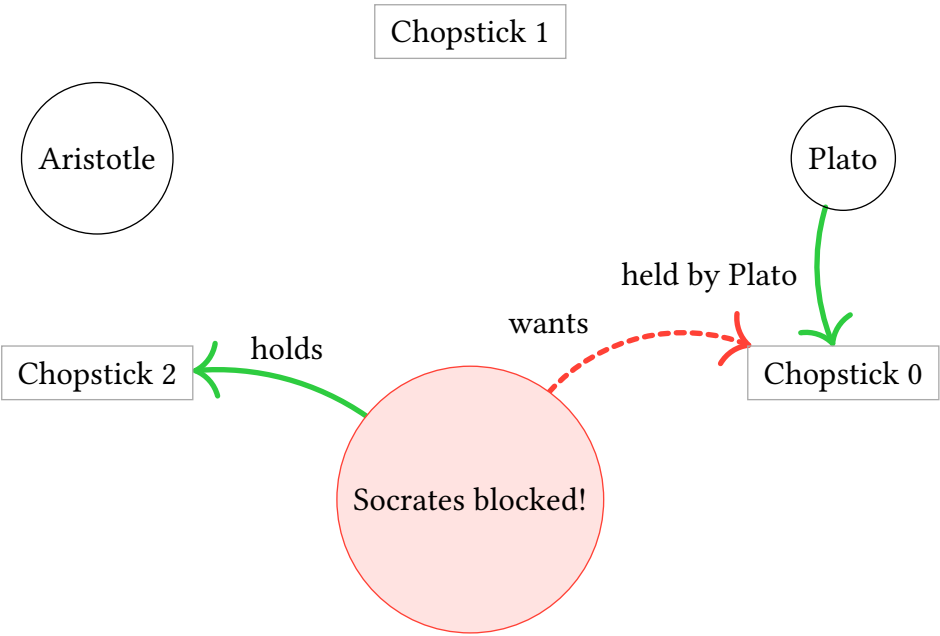
Which Coffman condition should we break?

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Same for Plato and Aristotle → circular wait → **Deadlock!**

Which Coffman condition should we break?

Circular wait: change lock order for one philosopher.

## 8.2. Assignment

Solve the “Dining Philosophers” problem with async tasks instead of threads.

- Revisit the standard thread-based version in `session6/examples/philosophers.rs`
- Complete exercise code in `session7/examples/philosophers.rs`.

Solutions are provided as `*-solution.rs` files.

## 8.3. Additional bonus

## 8. Exercise: async philosophers

Do we still need the asymmetry (the last philosopher switching hands)?

## 8.3. Additional bonus

Do we still need the asymmetry (the last philosopher switching hands)?

Yes, because all philosophers are still working in parallel in separate tasks, so we may have a deadlock.

How to solve this problem with only a single async task? Do we still have deadlocks



Do we still need the asymmetry (the last philosopher switching hands)?

Yes, because all philosophers are still working in parallel in separate tasks, so we may have a deadlock.

How to solve this problem with only a single async task? Do we still have deadlocks

No, since philosophers access sequentially.